INCH-POUND

A-A-59169
21 April 2004

COMMERCIAL ITEM DESCRIPTION
GENERATOR, FUNCTION
(0.1Hz to 15MHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a Function Generator which has a frequency range of 0.1Hz to 15MHz. This CID is meant as a minimum requirement for the Function Generator in which only those manufactures that meet or surpass the following requirements are supplied, per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample tested, if so requested, within the accuracies, limits, and specifications herein. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 4 in accordance with MIL-PRF-28800.

2.3. Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 8,000 hours of operation with a statistical certainty of 95%.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 6625

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2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.6 Power Requirements. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 200, 220, 240 volts AC (±10%) at line frequencies of 50Hz to 60Hz. Maximum power consumption shall not exceed 50 watts.

2.7 Operating Temperature. This instrument shall be capable of operating over a temperature range from -10°C to +40 °C.

2.8 Non-Operating (Storage) Temperature. This instrument shall be capable of being stored at temperatures ranging from –40 ^{0}C to +50 ^{0}C.

2.9. Physical Parameters. This instrument shall have the following parameters:
the weight shall not exceed a weight of 10 pounds and the nominal dimensions should be approximately 6 inches high x 12 inches wide x 15 inches deep.

2.10 PERFORMANCE REQUIREMENTS.

2.10.1 Frequency. The function generator frequency range shall be determined by the waveform selected.

2.10.2 Waveforms. The function generator shall provide a minimum of four (4) front panel selectable waveforms: sine, triangle, square and arbitrary.

2.10.2.1 Waveform Characteristics are as follows:

Sine Wave Characteristics:

Frequency Range: 0.1 mHz to 15 MHz

Resolution: 0.1 mHz (7 digits)

Accuracy: 20 ppm in one year

Harmonic Distortion:

< -60 dBc up to 20 kHz

< -50 dBc up to 1 MHz

< -35 dBc up to 10 MHz

Non-Harmonic Distortion: < -65 dBc up to 1 mHz

2.10.2.1.2 Square Wave Characteristics are as follows:

Frequency Range: 1 mHz to 15 MHz

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Rise/Fall Time: <25 ns

Resolution: 1 mHz (4 digits)

Accuracy: ± 1 digit of setting

2.10.2.1.3 Triangle Ramp Characteristics are as follows:

Frequency Range: 0.1 mHz to 100 kHz

Resolution: 0.1 mHz (7 digits)

Accuracy: 20 ppm for one year

Linearity: <0.1% up to 30 kHz

2.10.2.2 Output Characteristics are as follows:

Amplitude: 50 mVpp – 10 Vpp into a 50 ?

Amplitude Accuracy: < 2% at 1 kHz

Amplitude Flatness: ± 2 dB up to 200 kHz;

Amplitude Resolution: 3 digits or 1 mV

2.11.3 Operating Modes.

2.11.3.1 Continuous sweep. The selected waveform is output continuously at the programmed frequency. The function generator shall have a linear and a logarithmic sweep mode. The linear shall have a front-panel selectable sweep rate to 1:1000 and 1:100000 in the logarithmic mode.

2.11.3.2 Gated. The selected waveform is output continuously at he programmed frequency while the selected trigger signal is true.

2.11.3.3 Triggered Burst. Internal trigger generator rate of DC to 50 kHz; External trigger rate of DC to 1 MHz.

2.11.4 Adapters and peripherals. Unless otherwise specified, all signal input and output connectors shall be Type “BNC” female.

2.12 Remote Interfaces: RS-232 and IEEE-488.

2.13 Internal Cooling. The equipment shall be capable of operating continuously for twenty four hours at 55 ^{0}C. If forced air is used, it shall be as specified in MIL-PRF-28800. A thermal switch shall be provided which will interrupt circuit power to the function generator when the internal temperature rises to the degree of possible damage to the electronic components.

2.14.7 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

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3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability may compete by submission of bid samples. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met.

4.3 Exceeded Metric Products. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.3 IEEE standards can be obtained from IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331. Phone: 800-678-4333 (Toll-Free, USA & Canada), 732-981-0060 (Worldwide), Fax: 732-981-9667. Website: www.ieee.org

6.1.4 The Code of Federal Regulations (CRF) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-0918)

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